Jungtaek Kim

Personal Information

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Google Scholar:	https://scholar.google.com/citations?user=KXNUYWgAAAAJ

Work Experiences

Aug 10, 2026 – Present
Assistant Professor at **University of Arkansas**
Fayetteville, Arkansas, United States of America
I am affiliated with the Department of Electrical Engineering and Computer Science.

Nov 01, 2024 – Aug 07, 2026
Research Associate at **University of Wisconsin–Madison**
Madison, Wisconsin, United States of America
I worked on the reasoning and search capabilities of large language models with Prof. Kangwook Lee.

Jul 01, 2022 – Oct 31, 2024
Postdoctoral Associate at **University of Pittsburgh**
Pittsburgh, Pennsylvania, United States of America
This position was supported by the Center for Materials Data Science for Reliability and Degradation, which is the National Science Foundation's Industry-University Cooperative Research Center. I worked with Prof. Paul W. Leu and Prof. Oliver Hinder, and also collaborated with Prof. Satish Iyengar and Prof. Lucas Mentch.

Feb 16, 2022 – Jun 20, 2022
Postdoctoral Researcher
at **Pohang University of Science and Technology (POSTECH)**
Pohang, South Korea
This position was supported by the BK21 FOUR, funded by the Ministry of Education of Korea and National Research Foundation of Korea. It has been carried out in the group of Prof. Minsu Cho.

Jun 07, 2021 – Oct 01, 2021
Oct 30, 2020 – Feb 05, 2021
Research Intern at **Vector Institute**
Toronto, Ontario, Canada (Remote Research)
I have developed a Bayesian optimization-based LEGO assembly baseline and a reinforcement learning-based LEGO assembly method, dubbed Brick-by-Brick, during this internship. It has been carried out under the supervision of Prof. Graham W. Taylor.

Jan 02, 2018 – Apr 27, 2018
Research Engineering Intern at **SigOpt Inc.**
San Francisco, California, United States of America
I have implemented a Bayesian optimization method for transferring prior trials via multi-task Gaussian process regression. I worked with Dr. Michael McCourt. SigOpt Inc. has been acquired by Intel Corporation.

Oct 30, 2017 – Dec 29, 2017
Research Intern at **AITRICS**
Seoul, South Korea
I have implemented an attention model for a dataset of electronic health records and validated Bayesian optimization-based model tuning for clinical dataset. I worked with Dr. Saehoon Kim.

Jan 02, 2017 – Jan 26, 2017 Aug 01, 2016 – Aug 26, 2016	Research Intern at **Samsung Electronics** Hwaseong, South Korea I have studied, designed, and developed a trend classification framework for hundreds million sensors of semiconductor manufacturing system using convolutional neural networks, a density estimator for defects on semiconductors, and an integrated anomaly detection system for hundreds million sensors of semiconductor manufacturing system using convolutional recurrent neural networks and generative adversarial networks.

Education

Mar 01, 2015 – Feb 11, 2022	Ph.D. in Computer Science and Engineering at **Pohang University of Science and Technology (POSTECH)** Pohang, South Korea Supervisor: Prof. Seungjin Choi and Prof. Minsu Cho Dissertation: Efficient Bayesian Optimization: Algorithms, Approximation, and Regret Analysis Committee: Prof. Minsu Cho, Prof. Seungjin Choi, Prof. Dongwoo Kim, Dr. Michael McCourt, Prof. Jungseul Ok
Mar 01, 2010 – Feb 13, 2015	B.S. in Mechanical Engineering & Computer Science and Engineering at **Pohang University of Science and Technology (POSTECH)** Pohang, South Korea
Jul 07, 2014 – Aug 15, 2014	Summer Session at **University of California, Berkeley** Berkeley, California, United States of America
Mar 01, 2007 – Feb 28, 2010	High School Graduate at **Hansung Science High School** Seoul, South Korea

Research Interests

- Statistical machine learning
- Bayesian optimization
- Large language models
- Artificial intelligence for scientific discovery

Publications, Peer-Reviewed Presentations & E-Prints

(* and $^{+}$ indicate equal contribution by representing co-first and co-corresponding authors, respectively.)

E-prints

1. Hyun Jung Lee, **Jungtaek Kim**, Jongwon Jeong, Tae-Eui Kam, Donghyun Kim, and Yong Jae Lee. EMHO: Embodied agent harness optimization via experience traces. *arXiv e-prints*, arXiv:2610.08432, 2026.
2. Jaden Park*, **Jungtaek Kim***, Jongwon Jeong*, Robert D. Nowak, Kangwook Lee, and Yong Jae Lee. Exploration and exploitation errors are measurable for language model agents. *arXiv e-prints*, arXiv:2604.13151, 2026.
3. Jongwon Jeong, **Jungtaek Kim**, and Kangwook Lee. TAPE: Tool-guided adaptive planning and constrained execution in language model agents. *arXiv e-prints*, arXiv:2602.19633, 2026.
4. Sterling G. Baird, (skip 44 authors) **Jungtaek Kim**, (skip 78 authors) Yunheng Zou. Bayesian optimization hackathon for chemistry and materials. *ChemRxiv*, 2025.

5. **Jungtaek Kim**. Beyond regrets: Geometric metrics for Bayesian optimization. *arXiv e-prints*, arXiv:2401.01981, 2024.

6. **Jungtaek Kim** and Seungjin Choi. Practical Bayesian optimization with threshold-guided marginal likelihood maximization. *arXiv e-prints*, arXiv:1905.07540, 2019.

7. Minseop Park, **Jungtaek Kim**, Saehoon Kim, Yanbin Liu, and Seungjin Choi. MxML: Mixture of meta-learners for few-shot classification. *arXiv e-prints*, arXiv:1904.05658, 2019.

8. **Jungtaek Kim**, Saehoon Kim, and Seungjin Choi. Learning to warm-start Bayesian hyperparameter optimization. *arXiv e-prints*, arXiv:1710.06219, 2017.

Conferences

1. Yuchen Zeng*, Shuibai Zhang*, Wonjun Kang*, Shutong Wu, Lynnix Zou, Ying Fan, Heeju Kim, Ziqian Lin, **Jungtaek Kim**, Hyung Il Koo, Dimitris Papailiopoulos, and Kangwook Lee. ReJump: A tree-jump representation for analyzing and improving LLM reasoning. In *Proceedings of the Forty-Third International Conference on Machine Learning (ICML-2026)*, Seoul, South Korea, July 6–11, 2026.

2. Mijeong Kim, **Jungtaek Kim**, and Bohyung Han. GP-4DGS: Probabilistic 4D Gaussian splatting from monocular video via variational Gaussian processes. In *Proceedings of the 2026 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR-2026)*, Denver, Colorado, USA, June 3–7, 2026.

3. **Jungtaek Kim**. Density ratio estimation-based Bayesian optimization with semi-supervised learning. In *Proceedings of the Forty-Second International Conference on Machine Learning (ICML-2025)*, Vancouver, British Columbia, Canada, July 13–19, 2025.

4. Thomas Zeng, Shuibai Zhang, Shutong Wu, Christian Classen, Daewon Chae, Ethan Ewer, Minjae Lee, Heeju Kim, Wonjun Kang, Jackson Kunde, Ying Fan, **Jungtaek Kim**, Hyung Il Koo, Kannan Ramchandran, Dimitris Papailiopoulos, and Kangwook Lee. VersaPRM: Multi-domain process reward model via synthetic reasoning data. In *Proceedings of the Forty-Second International Conference on Machine Learning (ICML-2025)*, Vancouver, British Columbia, Canada, July 13–19, 2025. *Oral Presentation.*

5. Chaeyun Jang*, Hyungi Lee*, **Jungtaek Kim**$^{+}$, and Juho Lee^{+}. Model fusion through Bayesian optimization in language model fine-tuning. In *Advances in Neural Information Processing Systems 37 (NeurIPS-2024)*, Vancouver, British Columbia, Canada, December 10–15, 2024. *Spotlight Presentation.*

6. Hyunsoo Chung, **Jungtaek Kim**, Hyungeun Jo, and Hyungwon Choi. Exploiting preferences in loss functions for sequential recommendation via weak transitivity. In *Proceedings of the Thirty-Third ACM International Conference on Information and Knowledge Management (CIKM-2024)*, Boise, Idaho, USA, October 21–25, 2024. *Short Research Paper Track.*

7. Kwang-Sung Jun and **Jungtaek Kim**. Noise-adaptive confidence sets for linear bandits and application to Bayesian optimization. In *Proceedings of the Forty-First International Conference on Machine Learning (ICML-2024)*, Vienna, Austria, July 21–27, 2024.

8. **Jungtaek Kim**, Jeongbeen Yoon, and Minsu Cho. Generalized neural sorting networks with error-free differentiable swap functions. In *Proceedings of the Twelfth International Conference on Learning Representations (ICLR-2024)*, Vienna, Austria, May 7–11, 2024.

9. **Jungtaek Kim**, Mingxuan Li, Oliver Hinder, and Paul W. Leu. Datasets and benchmarks for nanophotonic structure and parametric design simulations. In *Advances in Neural Information Processing Systems 36 (NeurIPS-2023)*, New Orleans, Louisiana, USA, December 10–16, 2023. *Datasets and Benchmarks Track.*

10. Tackgeun You, Mijeong Kim, **Jungtaek Kim**, and Bohyung Han. Generative neural fields by mixtures of neural implicit functions. In *Advances in Neural Information Processing Systems 36 (NeurIPS-2023)*, New Orleans, Louisiana, USA, December 10–16, 2023.

11. **Jungtaek Kim**, Seungjin Choi^{+}, and Minsu Cho^{+}. Combinatorial Bayesian optimization with random mapping functions to convex polytopes. In *Proceedings of the Thirty-Eighth Conference on Uncertainty in Artificial Intelligence (UAI-2022)*, Eindhoven, the Netherlands, August 1–5, 2022.

12. Jinhwi Lee*, **Jungtaek Kim***, Hyunsoo Chung, Jaesik Park, and Minsu Cho. Learning to assemble geometric shapes. In *Proceedings of the Thirty-First International Joint Conference on Artificial Intelligence (IJCAI-2022)*, Vienna, Austria, July 23–29, 2022.

13. Rylee Thompson, Boris Knyazev, Elahe Ghalebi, **Jungtaek Kim**, and Graham W. Taylor. On evaluation metrics for graph generative models. In *Proceedings of the Tenth International Conference on Learning Representations (ICLR-2022)*, Virtual, April 25–29, 2022.

14. **Jungtaek Kim** and Seungjin Choi. On uncertainty estimation by tree-based surrogate models in sequential model-based optimization. In *Proceedings of the Twenty-Fifth International Conference on Artificial Intelligence and Statistics (AISTATS-2022)*, Virtual, March 28–30, 2022.

15. Hyunsoo Chung*, **Jungtaek Kim***, Boris Knyazev, Jinhwi Lee, Graham W. Taylor, Jaesik Park, and Minsu Cho. Brick-by-Brick: Combinatorial construction with deep reinforcement learning. In *Advances in Neural Information Processing Systems 34 (NeurIPS-2021)*, Virtual, December 6–14, 2021.

16. Juho Lee*, Yoonho Lee*, **Jungtaek Kim**, Eunho Yang, Sung Ju Hwang, and Yee Whye Teh. Bootstrapping neural processes. In *Advances in Neural Information Processing Systems 33 (NeurIPS-2020)*, Virtual, December 6–12, 2020.

17. **Jungtaek Kim** and Seungjin Choi. On local optimizers of acquisition functions in Bayesian optimization. In *Proceedings of the European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (ECML-PKDD-2020)*, Virtual, September 14–18, 2020.

18. Juho Lee, Yoonho Lee, **Jungtaek Kim**, Adam R. Kosiorek, Seungjin Choi, and Yee Whye Teh. Set Transformer: A framework for attention-based permutation-invariant neural networks. In *Proceedings of the Thirty-Sixth International Conference on Machine Learning (ICML-2019)*, Long Beach, California, USA, June 9–15, 2019.

19. Inhyuk Jo, **Jungtaek Kim**, Hyohyeong Kang, Yong-Deok Kim, and Seungjin Choi. Open set recognition by regularizing classifier with fake data generated by generative adversarial networks. In *Proceedings of the Forty-Third IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP-2018)*, Calgary, Alberta, Canada, April 15–20, 2018.

20. **Jungtaek Kim** and Seungjin Choi. Clustering-guided GP-UCB for Bayesian optimization. In *Proceedings of the Forty-Third IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP-2018)*, Calgary, Alberta, Canada, April 15–20, 2018.

21. Saehoon Kim, **Jungtaek Kim**, and Seungjin Choi. On the optimal bit complexity of circulant binary embedding. In *Proceedings of the Thirty-Second AAAI Conference on Artificial Intelligence (AAAI-2018)*, New Orleans, Louisiana, USA, February 2–7, 2018.

Journals

1. Anya Li, Abdullah Al Mahmood, **Jungtaek Kim**, Guangyong Li, and Paul W. Leu. Low-cost dual Fabry-Pérot cavities for high-purity, angle-independent structural color, *Journal of Optics*, 2026.

2. **Jungtaek Kim**, Thomas Zeng, Ziqian Lin, Minjae Lee, Chungpa Lee, Jy-yong Sohn, Hyung Il Koo, and Kangwook Lee. Transformers in the Dark: Navigating unknown search spaces via bandit feedback. *Transactions on Machine Learning Research*, 2026. *J2C Certification.*

3. Mingxuan Li*, **Jungtaek Kim***, and Paul W. Leu. JaxLayerLumos: A JAX-based differentiable optical and radio frequency simulator for multilayer structures. *Journal of Open Source Software*, vol. 10, no. 114, p. 8572, 2025.

4. Seokjun Ahn*, **Jungtaek Kim***, Minsu Cho, and Jaesik Park. Budget-aware sequential brick assembly with efficient constraint satisfaction. *Transactions on Machine Learning Research*, 2024.

5. Karinna Martin, Katie Shanks, Yipeng Liu, **Jungtaek Kim**, Sajad Haghanifar, Mehdi Zarei, Sooraj Sharma, and Paul W. Leu. Minimizing annual reflection loss in fixed-tilt photovoltaic modules using graded refractive index (GRIN) anti-reflective glass. *Solar Energy*, vol. 272, p. 112424, 2024.

6. Mehdi Zarei, Mingxuan Li, Elizabeth E. Medvedeva, Sooraj Sharma, **Jungtaek Kim**, Zefan Shao, S. Brett Walker, Melbs LeMieux, Qihan Liu, and Paul W. Leu. Flexible embedded metal meshes by sputter-free crack lithography for transparent electrodes and electromagnetic interference shielding. *ACS Applied Materials & Interfaces*, vol. 16, no. 5, pp. 6382–6393, 2024.

7. **Jungtaek Kim**, Mingxuan Li, Yirong Li, Andrés Gómez, Oliver Hinder, and Paul W. Leu. Multi-BOWS: Multi-fidelity multi-objective Bayesian optimization with warm starts for nanophotonic structure design. *Digital Discovery*, vol. 3, no. 2, pp. 381–391, 2024.

8. **Jungtaek Kim** and Seungjin Choi. BayesO: A Bayesian optimization framework in Python. *Journal of Open Source Software*, vol. 8, no. 90, p. 5320, 2023.

9. **Jungtaek Kim**, Michael McCourt, Tackgeun You, Saehoon Kim, and Seungjin Choi. Bayesian optimization with approximate set kernels. *Machine Learning*, vol. 110, no. 5, pp. 857–879, 2021.

Peer-Reviewed Workshops

1. Hyun Jung Lee, **Jungtaek Kim**, Jongwon Jeong, Tae-Eui Kam, Donghyun Kim, and Yong Jae Lee. Harness optimization for embodied agents via experience traces. *NeurIPS Workshop on Self-Evolving Diversity-Driven Search for Robust AI Systems (EvoRobust-2026)*, Seoul, South Korea, Sydney, Australia, December 11, 2026.

2. Jaden Park*, **Jungtaek Kim***, Jongwon Jeong*, Robert D. Nowak, Kangwook Lee, and Yong Jae Lee. Exploration and exploitation errors are measurable for language model agents. *ICML Workshop on Failure Modes in Agentic AI: Reproducible Triggers, Trace Diagnostics, and Verified Fixes (FAGEN-2026)*, Seoul, South Korea, July 10, 2026.

3. Jongwon Jeong, **Jungtaek Kim**, and Kangwook Lee. TAPE: Tool-guided adaptive planning and constrained execution in language model agents. *ICLR Workshop on Agentic AI in the Wild: From Hallucinations to Reliable Autonomy (AAIW-2026)*, Rio de Janeiro, Brazil, April 27, 2026.

4. **Jungtaek Kim**, Ziqian Lin, Thomas Zeng, Minjae Lee, Chungpa Lee, Jy-yong Sohn, Hyung Il Koo, and Kangwook Lee. Transformers in the Dark: Navigating unknown search spaces via noisy feedback. *NeurIPS Workshop on What Can('t) Transformers Do? (WCTD-2025)*, San Diego, California, USA, December 7, 2025.

5. Mingxuan Li, **Jungtaek Kim**, Oliver Hinder, and Paul W. Leu. CHROMA: Conversational human-readable optical multilayer assembly for natural language-driven inverse design of structural coloration. *NeurIPS Workshop on AI for Accelerated Materials Design (AI4Mat-2025)*, San Diego, California, USA, December 6, 2025.

6. Mingxuan Li, **Jungtaek Kim**, and Paul W. Leu. Discovering multi-layer films for electromagnetic interference shielding and passive cooling with multi-objective active learning. *NeurIPS Workshop on AI for Accelerated Materials Design (AI4Mat-2024)*, Vancouver, British Columbia, Canada, December 14, 2024.

7. Chaeyun Jang, **Jungtaek Kim**, Hyungi Lee, and Juho Lee. Model fusion through Bayesian optimization in language model fine-tuning. *NeurIPS Workshop on Efficient Natural Language and Speech Processing (ENLSP-2023)*, New Orleans, Louisiana, USA, December 16, 2023.

8. Hyunsoo Chung and **Jungtaek Kim**. Leveraging uniformity of normalized embeddings for sequential recommendation. *NeurIPS Workshop on Self-Supervised Learning - Theory and Practice (SSL-TP-2023)*, New Orleans, Louisiana, USA, December 16, 2023.

9. **Jungtaek Kim**, Hyunsoo Chung, Jinhwi Lee, Minsu Cho, and Jaesik Park. Combinatorial 3D shape generation via sequential assembly. *NeurIPS Workshop on Machine Learning for Engineering Modeling, Simulation, and Design (ML4Eng-2020)*, Virtual, December 12, 2020.

10. Jinhwi Lee*, **Jungtaek Kim***, Hyunsoo Chung, Jaesik Park, and Minsu Cho. Fragment relation networks for geometric shape assembly. *NeurIPS Workshop on Learning Meets Combinatorial Algorithms (LMCA-2020)*, Virtual, December 12, 2020.

11. **Jungtaek Kim**, Michael McCourt, Tackgeun You, Saehoon Kim, and Seungjin Choi. Bayesian optimization over sets. *ICML Workshop on Automated Machine Learning (AutoML-2019)*, Long Beach, California, USA, June 14, 2019.

12. Minseop Park, Saehoon Kim, **Jungtaek Kim**, Yanbin Liu, and Seungjin Choi. TAEML: Task-adaptive ensemble of meta-learners. *NeurIPS Workshop on Meta-Learning (MetaLearn-2018)*, Montreal, Quebec, Canada, December 8, 2018.

13. **Jungtaek Kim** and Seungjin Choi. Automated machine learning for soft voting in an ensemble of tree-based classifiers. *ICML Workshop on Automatic Machine Learning (AutoML-2018)*, Stockholm, Sweden, July 14, 2018.

14. **Jungtaek Kim**, Saehoon Kim, and Seungjin Choi. Learning to transfer initializations for Bayesian hyperparameter optimization. *NeurIPS Workshop on Bayesian Optimization (BayesOpt-2017)*, Long Beach, California, USA, December 9, 2017.

15. **Jungtaek Kim**, Jongheon Jeong, and Seungjin Choi. AutoML Challenge: AutoML framework using random space partitioning optimizer. *ICML Workshop on Automatic Machine Learning (AutoML-2016)*, New York, New York, USA, June 24, 2016.

Selected Software

- BayesO: A Bayesian optimization framework in Python.
  - BayesO, BayesO Benchmarks, Batch BayesO, BayesO Metrics
- Combinatorial 3D Shape Dataset: Dataset for the sequences of volumetric primitives, i.e., LEGO bricks, composed of 406 instances of 14 classes.
- Nanophotonic Structures: Datasets and benchmarks for nanophotonic structure and parametric design simulations.
- JaxLayerLumos: A JAX-based differentiable optical and radio frequency simulator for multilayer structures.

Funding

1. AI Working Group: Physical AI and AI for Physical Systems (PI: Jungtaek Kim, Co-PI: Hasan Shahriar and Chase Rainwater), PI, College of Engineering, University of Arkansas, $10,000, October 2026 – July 2027.
2. Empirical Analysis of the Reasoning Abilities of Large Language Models Through the Lens of Chain-of-Thought and Tree-of-Thought (CIS251382, PI: Jungtaek Kim), PI, NSF ACCESS, 750,000 ACCESS credits, December 2025 – December 2026.
3. Understanding the Importance of Reasoning Ability in Large Language Models for Scientific Discovery (PI: Jungtaek Kim), PI, Google Cloud Research Credits Program, $5,000, March 2025 – March 2026.
4. Machine Learning Methods for Optimizing and Innovating Structural Color Paints and Coatings (PI: Paul W. Leu, Oliver Hinder, and Jungtaek Kim), PI, MDS-Rely, $48,220.4, January 2024 – August 2025.

Honors and Awards

Jun 09, 2019	ICML Travel Award for ICML-2019
Jun 06, 2018	2nd place in AutoML Challenge 2018 (PAKDD-2018 Data Competition)
Apr 15, 2018	IEEE Signal Processing Society Travel Grant for ICASSP-2018
Nov 18, 2016	Best Paper Runner-Up Award (supported by LG U+) in 2016 Fall Conference of Korea Business Intelligence Data Mining Society
Jun 24, 2016	3rd place in AutoML5 Phase of AutoML Challenge
Jul 03, 2014	Software Maestro (organized by Ministry of Science, ICT and Future Planning, South Korea & National IT Promotion Agency, South Korea)
Mar 02, 2010	Presidential Science Scholarship (awarded by the President of South Korea)

Professional Services

Area Chairs

- AISTATS (2027), ICLR (2027)

Program Committee Members & Reviewers

- NeurIPS (2026, 2025, 2024, 2023, 2022, 2021, 2020, 2019), ICML (2026, 2025, 2024, 2023, 2022, 2021, 2020, 2019), AISTATS (2026, 2025, 2024, 2023, 2022, 2021), UAI (2025, 2024, 2023, 2022, 2020, 2019), ICLR (2027, 2026, 2025, 2024, 2023, 2022), ECML-PKDD (2020), ACML (2022, 2021, 2020, 2019, 2018), AutoML-Conf (2022), CVPR (2026, 2023), ICCV (2025), AAAI (2023, 2022, 2021, 2020), IJCAI (2022, 2021), IJCNN (2019, 2018, 2017)
- Bayesian Analysis, Transactions on Machine Learning Research, Machine Learning, Journal of Open Source Software, Journal of Artificial Intelligence Research, IEEE Transactions on Image Processing, International Journal of Computer Vision
- ICLRW AAIW (2026), NeurIPSW WCTD (2025), NeurIPSW SSL-TP (2023), ICLRW NAS (2020), ICMLW AutoML (2021, 2020, 2019, 2018)
- Directorate for Geosciences (GEO), U.S. NSF (2025)

Talks & Non-Peer Reviewed Presentations

- University of Minnesota (September 18, 2026), MOGAM Institute for Biomedical Research (September 5, 2023), Sequential Decision Making for Optimization, Learning and Search, The Tenth International Congress on Industrial and Applied Mathematics (August 24, 2023), BIRS Workshop on 3D Generative Models (July 12, 2023), MDS-Rely Spring 2023 Meeting (April 17, 2023), Computer Science and Engineering Department Seminar, Washington University in St. Louis (May 5, 2023), Technical Seminar, MDS-Rely (September 15, 2022), The Eighth Annual Data Science in Engineering and Life Sciences Symposium (August 8, 2022), Graph Representation Learning Reading Group, Mila (April 14, 2022), KIRO (June 18, 2021), LIM Lab., UNIST (April 22, 2021), Vector Institute Research Symposium (February 16, 2021), Tutorial at KSC-2020 (December 23, 2020), COSEAL-2019 (August 26, 2019), AI Korea 2019 (July 26, 2019), Semiconductor Research Center, Samsung Electronics (June 28, 2019), Samsung Advanced Institute of Technology, Samsung Electronics (December 13, 2018), Naver Corporation (June 12, 2018), SigOpt Inc. (January 04, 2018), Yonsei University Health System (December 15, 2017), Software R&D Center, Samsung Electronics (October 11, 2017), Manufacturing Technology Center, Samsung Electronics (April 14, 2017), Chemical Engineering Department, Yeungnam University (October 05, 2016), The Fifteenth KYUTECH-POSTECH Joint Workshop on Neuroinformatics (August 23, 2016), XBrain Inc. (January 03, 2016)

Teaching

Courses

- Fall 2026, Artificial Intelligence @ University of Arkansas
- Spring 2022, Recent Trends in Machine Learning: A Large-Scale Perspective @ POSTECH

Teaching Assistants

- Artificial Intelligence, Probabilistic Graphical Models, Automata & Formal Languages, Machine Learning
- Hyundai Steel Company ML Course, SK Hynix ML Course, Samsung Electronics Device Solutions Business ML Course, POSCO Group ML Course

Languages

Korean: Native
English: Full professional proficiency